AGATHE SADEGHI

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EXPERIENCE

Uniswap Labs, New York City, NY **Sep 2025 - Present**

Quant Research Fellow

- Developed statistical and machine learning models for pattern recognition and classification of trader and liquidity provider (LP) behavior, including statistical arbitrageurs and active/passive LPs, and engineered predictive features driving market making strategy and execution alpha.
- Researched market microstructure across decentralized exchanges using high frequency, transaction level data to model adverse selection, flow toxicity, and execution quality.
- Architected scalable data pipelines for large scale swap, liquidity, and pricing datasets, and built robust backtesting engines to analyze market patterns and quantify volume migration and profitability across fragmented liquidity venues (AMMs, PropAMMs and RFQs).
- Developed Smart Order Routing (SOR) optimization algorithms leveraging synthetic Monte Carlo simulations and empirical trade data along with fee-level robustness analysis to dynamically adjust fee strategies and maximize price improvement without A/B testing.

Bloomberg LP, New York City, NY **May - Aug 2023**

Quantitative Research Intern

- Developed a Kernel-Neural-Net causal discovery model for nonstationary financial time series to isolate contemporaneous and lagged dependencies for alpha signal generation and statistical market pattern detection.
- Engineered features, trained models, and evaluated predictive relationships using Python (NumPy, Pandas, PyTorch) and GPU acceleration; visualized results with Matplotlib and interactive Jupyter workflows.
- Conducted robust backtesting of trading signals on fundamental data, explicitly addressing distribution shifts and nonstationarity to ensure signal efficacy and minimize overfitting across changing market regimes.

Bloomberg LP, New York City, NY **June - Aug 2022**

Quant Analytics Intern

- Modeled directed network topologies using graph theory (NetworkX) to analyze partial correlations, conditional expectations, and complex systemic dependencies within Credit Default Swap (CDS) markets.
- Improved corporate default risk (DRSK) model by implementing stochastic Monte Carlo simulations, framing equity as a down-and-out barrier option on firm assets utilizing structural credit models.

Stevens Institute of Technology, Hoboken, NJ **Sep 2020 - May 2025**

Instructor

- Instructed graduate students on quantitative data extraction and analysis using Bloomberg Terminal (BQL) and APIs for Thomson Reuters, Capital IQ, and WRDS.

EDUCATION

Stevens Institute of Technology, Hoboken, NJ **Aug 2025**

Doctor of Philosophy in Financial Engineering GPA: 3.90/4.00

- Dissertation Topic: Contagion Dynamics in Traditional and Decentralized Finance
- Advisor: Prof. Zachary Feinstein

Columbia University, New York City, NY **Jan - May 2025**

Visiting PhD Student

- Industrial Engineering and Operations Research Department, Center for Digital Finance and Technologies.
- Mentor: Prof. Agostino Capponi

Stevens Institute of Technology, Hoboken, NJ **Dec 2020**

Master of Science in Financial Engineering GPA: 3.92/4.00

- Awarded the competitive Provost's Master Fellowship and the esteemed Jérôme Lohez Scholarship
- Thesis Topic: Corporate Bond Yield Predictability: A Statistical and Machine Learning Approach

Iran University of Science and Technology, Tehran, Iran

Master of Science in Systems Engineering **Dec 2020**

Bachelor of Science in Industrial Engineering **Sep 2018**

SELECTED PUBLICATIONS, PREPRINTS AND WORKING PAPERS

- Sadeghi, A.; Feinstein, Z. Liquidation Dynamics in DeFi and the Role of Transaction Fees. Submitted to Digital Finance (2026). Link
- Sadeghi, A.; Moallemi, C.; Liu, Dingyue; Zhu, Brian Z. Not All LPs Are Equal: The Active-Passive Gap in Automated Market Maker Liquidity Provision (2026).
- Sadeghi A; Konstantinov G. "Unsupervised Learning: Network Theory" Chapter in "AI in Asset Management: Tools, Applications, and Frontiers". CFA Institute (2025). Link
- Sadeghi, A.; Feinstein, Z. Statistical Validation of Contagion Centrality in Financial Networks. Accepted in SIAM FM (2025). Link
- Sadeghi, A.; Simonian, J. Deciphering the Causal Drivers of Bond Indices: A Double Machine Learning Approach. JFDS (2025). Link
- Sadeghi, A.; Feinstein, Z. Network-Based Contagion Risk Factor for Joint VaR and Expected Shortfall Prediction (2025).
- Sadeghi, A.; Gopal, A.; Fesanghary, M. Causal Discovery from Nonstationary Time Series. IJDSA (2024). Link.

ADDITIONAL INFORMATION

- **Skills:** Python, SQL, Git, BigQuery, DataForm, Hex, Dune, R, Bloomberg Terminal, BQL, MATLAB, LaTex
- **Workshops:** Columbia CryptoEconomics Workshop - Dec 2025 (Columbia University), CBER Crafting the Cryptoeconomy Conference - Oct 2025 (Columbia University), Cryptoeconomics Summer School - Sept 2025 (Columbia University), Crypto Academic Camp during Edge Esmeralda - June 2026 (Healdsburg, CA)
- **Languages:** English (proficient), Armenian (native), Persian (native), Arabic (intermediate), German (basic)
- **Hobbies:** Tennis and USTA Official, Mountaineering, Backgammon